CIE AS PHYSICS (9702) • QUESTION PAPER

1.1 Physical Quantities

Syllabus Section: 1. Physical Quantities & Units **Paper Type:** Multiple Choice (Paper 1) **Total:** 17 Questions (17 Marks)

Question 1 Easy [1 mark]

Which definition is correct and uses only quantities rather than units?

A Density is mass per cubic metre

B Potential difference is energy per unit current

C Pressure is force per unit area

D Speed is distance travelled per second

Question 2 Easy [1 mark]

What is a reasonable estimate for the gravitational force acting on an adult person standing on the Earth?

A 50 N

B 300 N

C 400 N

D 700 N

Question 3 Easy [1 mark]

A physical quantity has the symbol V and the unit m^3.

Which physical quantity does V represent?

A Potential difference

B Velocity

C Pressure

D Volume

Question 4 Easy [1 mark]

What is a reasonable estimate of the number of seconds in a year?

A 5×10^5 s

B 1×10^6 s

C 3×10^7 s

D 7×10^8 s

Question 5 Easy [1 mark]

A physical quantity has the unit m s^{-2} and the symbol a.

What is the correct name for this quantity?

A Activity of a radioactive source

B Amplitude

C Avogadro constant

D Acceleration

Question 6 Easy [1 mark]

Which of the following must all physical quantities have?

A Direction and magnitude

B Direction and a unit

C Magnitude and a prefix

D Magnitude and a unit

Question 7 Medium [1 mark]

A student creates a table to show reasonable estimates of some physical quantities.

Which row is a reasonable estimate?

	quantity	value
A	frequency of a radio wave	500 pHz
B	mass of an atom	500 μg
C	the Young modulus of a metal	500 kPa
D	wavelength of green light	500 nm

A

B

C

D

Question 8 Medium [1 mark]

What is a reasonable estimate for the volume of 1 ml of water?

A $0.1\ cm^3$

B $1\ cm^3$

C $10\ cm^3$

D $100\ cm^3$

Question 9 Medium [1 mark]

Which row shows what all physical quantities must have?

	magnitude	direction	unit
A	✗	✗	✓
B	✓	✗	✓
C	✓	✓	✗
D	✓	✓	✓

A

B

C

D

Question 10 Medium [1 mark]

An adult male stands with a litre carton of milk balanced on his head.

What is the best estimate of the gravitational potential energy of the milk carton?

A 2 J

B 20 J

C 200 J

D 2000 J

Question 11 Medium [1 mark]

Which of the following is a reasonable estimate for the density of a sugar cube?

A $0.01\ \text{kg m}^{-3}$

B $0.1\ \text{kg m}^{-3}$

C $10\ \text{kg m}^{-3}$

D $1000\ \text{kg m}^{-3}$

Question 12 Hard [1 mark]

Which row shows a reasonable estimate?

	quantity	value
A	kinetic energy of a bus travelling on an expressway	30 000 J
B	power of a domestic light bulb	300 W
C	temperature of a hot oven	300 K
D	volume of air in a car tyre	$0.03\ \text{m}^3$

A

B

C

D

Question 13 Hard [1 mark]

Which statement is **incorrect** by a factor of 100 or more?

A Atmospheric pressure is about 1×10^5 Pa

B Light takes 5×10^2 s to reach us from the Sun

C The frequency of ultraviolet light is 3×10^{12} Hz

D The life-span of a man is about 2×10^9 s

Question 14 Hard [1 mark]

An adult person jumps from a wall which is 2 m high. They land without rebounding.

What is the best estimate for the change in momentum of the person?

A 4×10^0 kg m s^{-1}

B 4×10^1 kg m s^{-1}

C 4×10^2 kg m s^{-1}

D 4×10^3 kg m s^{-1}

Question 15 Hard [1 mark]

The diameter of a proton is approximately 1×10^{-15} m.

Which estimate is reasonable for the time taken for light to cross the nucleus of a hydrogen atom?

A 3×10^{-28} s

B 3×10^{-24} s

C 3×10^{-20} s

D 3×10^{-16} s

Question 16 Hard [1 mark]

The bottom of the Mariana Trench is estimated to be approximately 11 000 m below sea level.

What is the best estimate of the total pressure at the bottom of the Mariana Trench?

A 1×10^{6} Pa

B 1×10^{7} Pa

C 1×10^{8} Pa

D 1×10^{9} Pa

Question 17 Hard [1 mark]

A satellite orbits Earth at an altitude of 400 km above the Earth's surface.

What is a reasonable estimate of the total distance travelled by the satellite in one complete orbit?

A 4×10^5 m

B 4×10^6 m

C 4×10^7 m

D 4×10^8 m